

SIDDHESH SANTOSH SHINGATE

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EDUCATION

Northeastern University, Master of Science in Robotics Engineering (CGPA: 3.70/4.00)	Sept 2024 – May 2026
Manipal University Jaipur, B.Tech in Mechatronics Engineering (CGPA: 3.41/4.00)	Jul 2019 – Jun 2023

WORK EXPERIENCE

Robotics Research Assistant Field Robotics Lab, Northeastern University	May 2025 – Present
Master’s Project: Designed, deployed, and validated an asynchronous stereo camera calibration system without hardware synchronization, jointly estimating temporal offset and SE(3) extrinsic in continuous time; recovered a 48.4 ms offset, 223 mm baseline (50 ms, 221 mm GT) with under 0.3 px reprojection error. - Conducted failure analysis on VI SLAM system (ORB-SLAM3) deployed on quadraped, diagnosing 99.4% feature tracking loss by profiling C++ front-end source code; traced root cause to gait-induced pitch/roll oscillations (8-12 deg) causing simultaneous BRIEF descriptor breakdown and constant-velocity motion model prediction failure. - Developed ROS2-based data extraction and preprocessing pipelines for synchronized LiDAR, stereo camera data from MCAP logs, enabling sensor calibration, perception evaluation, and offline analysis of field robotics experiments.	
Graduate Teaching Assistant Robotics Sensing and Navigation, Northeastern University	Jan 2026 – May 2026
Mentored 90+ students in building and debugging robot perception, state estimation pipelines in Python, C++, and ROS2, including GNSS/IMU/camera integration, calibration workflows, and real-time debugging; maintained ROS2 course infrastructure and Git repository.	

PROJECTS

Multi-View 3D Reconstruction (Structure from Motion)
- Built a full 3D reconstruction system from scratch using SIFT feature extraction, RANSAC geometric verification, essential matrix estimation, camera projection models and triangulation to recover globally consistent camera motion and 3D structure from 20+ images. - Integrated GTSAM-based factor graph bundle adjustment across frame transformations, reducing mean reprojection error from 0.76 px to 0.39 px (49% reduction) and improving global reconstruction consistency.
Robust Vehicle Localization with EKF in GPS-Degraded Environments
- Developed a real-time multi-sensor state estimation pipeline using an Extended Kalman Filter, fusing 200 Hz IMU and 10 Hz GPS measurements to provide continuous localization during GPS outages, reducing pure inertial drift from 14 km to 154 m over a 16.5 km route. - Tuned sensor noise models from real-world data for fault-tolerant localization under degraded signal conditions.
Underwater Photomosaic with Pose Graph SLAM
Eliminated 58px accumulated drift across a 28-frame Skerki Bank shipwreck in lawn mower pattern (500m depth) by building a GTSAM pose graph SLAM system with 25 loop closures; used SIFT/RANSAC homography inlier-count-weighted edges to reject false loop closures, reducing total trajectory uncertainty by 34%.
G1Reach: End-Effector Control via Reinforcement Learning
- Developed a reinforcement learning policy for autonomous manipulation on a 29-DOF humanoid robot (Unitree G1) arm using PPO and GAE, achieving 80% task success rate and 8.4 cm end-effector accuracy on 3D reaching tasks. - Built a MuJoCo simulation environment with domain randomization, 225-dimensional proprioceptive observations, and PD low-level motor control, achieving 3,666 FPS for rapid RL policy iteration.
Cartesian-Frenet Nonlinear Model Predictive Controller for Autonomous Navigation
- Developed a nonlinear model predictive control (MPC) framework for autonomous trajectory generation, tracking, and collision avoidance using CasADi/IPOPT, executing feasible plans at 10 Hz with a 78 ms solver runtime. - Validated navigation performance over a 112 m trajectory including sharp turns and U-turns, achieving 17 cm obstacle clearance and 0.5 m RMS tracking error under 0.3 m/s^2 acceleration constraints.

TECHNICAL SKILLS

Languages & Libraries: Python, C++, OpenCV, GTSAM, PyTorch, Open3D, NumPy, Gymnasium
Robotics Systems: ROS2, Linux, Git, Docker, Gazebo, Isaac Sim, MuJoCo, Foxglove, Kalibr, AprilTags
Core Areas: Calibration, Localization, State Estimation, Sensor Fusion, SLAM, Perception, Motion Planning, Trajectory Optimization, Nonlinear Optimization, Robot Kinematics, Dynamics, Hardware-Software Integration, Failure Analysis, Reinforcement Learning, Deep Learning